

PARTS LIST

Semiconductors:

T1, T2 - BC547 npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1-R5 - 12-kilo-ohm

R6 - 330-ohm

R7, R8 - 22-kilo-ohm

Capacitors:

C1 - 100 μ F, 10V electrolytic

C2 - 100nF ceramic disk

C3 - 470pF ceramic disk

Miscellaneous:

MOD1 - ESP8266-12E module

MOD2 - FT232 module

MOD3 - 3.3V module

S1, S2 - Tactile switch

(MOD1), FT232 module (MOD2), 3.3V module (MOD3), two switches (S1 and S2), two BC547 transistors (T1 and T2) and a few other components.

Automatic ESP-12E/F programmer circuit is similar to the manual programmer discussed above. RTS signal is used for resetting ESP8266, while DTR signal is used for putting the chip in flash mode. T1 and T2 are used so that ESP8266 is not reset

when both DTR and RTS signals are low. On detecting upload command, ESP module is automatically put in flash mode, and a new sketch can be uploaded to it seamlessly. In the absence of upload command, ESP-12E/F starts in normal mode, and the programmed ESP-12E/F board can be tested or used in an application.

To test ESP-12E/F board, a simple LED blinking application is presented in this article. The software code (NewBlink.ino) is written in Arduino programming language.

How to use the programmer

The programmer can be built on a general-purpose PCB, as shown in Fig. 7, or on the designed PCB, as shown in Figs 8 and 9.

Female Berg strip connectors are used for plugging in ESP8266 chip and FTDI board (FT232 module) on the programmer board. Mount the modules and connectors on the PCB. Make sure that the modules are plugged in the correct orientation.

Now, connect FT232 module to the computer via a USB cable. Open Arduino IDE and write your code, or open the existing code stored in your computer. Here, you just need to open the code (NewBlink.ino) of the LED blinking application.

Select the correct Com port in Arduino IDE and set the board as NodeMCU 1.0 (ESP-12E Module). Check that the upload speed is 115200. Compile and upload as you would normally do using NodeMCU. Built-in LED (on GPIO 2) on ESP-12E/F board will start blinking on successful upload.

The programmed ESP-12E/F module can be unplugged from the programmer and shifted to your project-specific board. You now have a cheaper and smaller version of NodeMCU ready for deployment. **EFY**



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